

QMS 101 Introductory Statistics

Tutorial Sheet — Topic VI: Probability Theory

Dr. Anna Fome

2026-06-19

Instructions: Attempt every question on your own first, writing out your full working. Only check the worked solution after you have tried. Mathematics is learned by doing, not by reading.

Section A: Basic Probability

Question 1

A bag contains **20 fruits**: 8 mangoes, 7 oranges, and 5 bananas. One fruit is picked at random.

(a) Write down the sample space.

(b) Find:

- (i) $P(\text{mango})$
- (ii) $P(\text{not a mango})$
- (iii) $P(\text{orange or banana})$

(c) Verify that your answers to (b)(i) and (b)(ii) sum to 1.

(d) A student says: “*There are 3 fruit types, so $P(\text{mango}) = \frac{1}{3}$.*” Explain why this is incorrect.

Question 2

Out of **250 past growing seasons** in a district, **175 seasons** had adequate rainfall.

(a) Using the empirical (relative frequency) approach, find the probability that the next season has adequate rainfall.

(b) State **one** assumption this approach relies on.

(c) An agricultural officer with 30 years of experience says: “*I believe there is roughly an 80% chance of adequate rainfall this year, based on the early signs I am seeing.*” Which of the three approaches to probability is this — classical, empirical, or subjective? Explain.

Section B: Types of Events

Question 3

A poultry farmer classifies each chicken in a flock as either **healthy** or **sick**. Separately, each chicken is also classified by **age**: chick, grower, or layer.

- (a) Are “healthy” and “sick” mutually exclusive? Explain.
- (b) Are “chick” and “grower” mutually exclusive? Explain.
- (c) Suppose 200 chickens are surveyed: 40 are chicks, 90 are growers, and 70 are layers. Are these three age categories collectively exhaustive? Justify your answer using the numbers given.
- (d) A farmer randomly inspects one chicken from the Northern pen and, separately, one chicken from a Southern pen 50 km away. Are the health statuses of these two chickens likely to be independent or dependent? Explain your reasoning.

Question 4

A box contains **6 light bulbs**: 2 are faulty and 4 are working. Two bulbs are selected one after another **without replacement**.

- (a) Find $P(\text{first bulb is faulty})$.
- (b) Find $P(\text{second bulb is faulty} \mid \text{first bulb was faulty})$.
- (c) Are the two draws independent or dependent? Justify your answer using your results from (a) and (b).

Section C: Addition Rule and Multiplication Rule

Question 5

A district survey of **80 farms** found: 35 use **certified seeds (C)**, 28 use **organic manure (O)**, and 12 use **both**.

- (a) Are C and O mutually exclusive? Justify with numbers.
- (b) Find $P(C \cup O)$ — the probability a farm uses certified seeds or organic manure (or both). Show the addition rule clearly.
- (c) Find the probability that a farm uses **neither** certified seeds nor organic manure.
- (d) Find the probability that a farm uses certified seeds **only** (not organic manure).

Question 6

Two independent pest-control treatments are applied to two different farms.

- Farm 1 has a 90% chance the treatment succeeds: $P(\text{success}_1) = 0.90$
- Farm 2 has an 85% chance the treatment succeeds: $P(\text{success}_2) = 0.85$

- (a) Find the probability that **both** treatments succeed.
- (b) Find the probability that **both** treatments fail.
- (c) Find the probability that **at least one** treatment succeeds.

(d) Find the probability that **exactly one** treatment succeeds.

Question 7

A sack contains **12 bags of rice**: 4 are understocked (underweight) and 8 are correctly filled. Two bags are drawn **without replacement**.

(a) Find $P(\text{both bags are understocked})$.

(b) Find $P(\text{both bags are correctly filled})$.

(c) Find $P(\text{at least one bag is understocked})$.

(d) Compare your answer in (a) to what the answer would be **with replacement**. Which is larger, and why?

Section D: Mixed Application Problems

Question 8

A cooperative has **100 members**. Of these, 55 attended a recent training workshop (T), and 40 members reported using improved storage techniques (S). 25 members did **both** — attended training and use improved storage.

(a) Construct a two-way (contingency) table summarising this information, including row and column totals.

(b) Find $P(T)$, $P(S)$, and $P(T \cap S)$.

(c) Find $P(T \cup S)$.

(d) Find the probability that a randomly selected member did **neither** attend training **nor** use improved storage.

(e) Two members are selected at random, **with replacement**. Find the probability that both attended training.

Question 9

A district agricultural office must:

1. Elect a **Chairperson, Secretary, and Treasurer** from **10 members** (three distinct roles).
2. Separately form an unranked **task force of 4** from the same 10 members.

(a) Calculate the number of ways to fill the three officer positions.

(b) Calculate the number of possible task forces of 4.

(c) A box contains 10 tickets, one for each of the 10 members. Two tickets are drawn at random without replacement to award a “Most Improved Farmer” prize and a “Community Service” prize (two distinct prizes — the same person cannot win both). How many different outcomes are possible?

(d) From the same 10 members, one farm is randomly chosen to host a field visit. There is a 30% chance the visit happens on a rainy day, and a 70% chance it happens on a dry day. If it is rainy, there is a 0.80 probability the visit is postponed. If it is dry, there is only a 0.05 probability the visit is postponed. Find the overall probability that the visit is postponed. (*Hint: this combines ideas from Topic VI’s multiplication and addition rules — break into rainy and dry scenarios.*)

Section E: Expected Value

Question 10

A trader buys a perishable batch of tomatoes for resale. Based on past experience, the trader's profit depends on demand:

Demand Level	Probability	Profit (TZS '000)
High	0.30	60
Medium	0.45	35
Low	0.25	-10

- (a) Verify that the probabilities sum to 1.
- (b) Calculate the expected profit $E(X)$. Show a full working table.
- (c) Calculate $E(X^2)$.
- (d) Calculate $\text{Var}(X)$ and $\text{SD}(X)$.
- (e) A second trader has a guaranteed profit of TZS 30,000 with certainty (no risk at all). Compare the two traders using expected value and standard deviation, and advise a risk-averse investor which option to choose.

Question 11

A motorbike-taxi (bajaji) operator is deciding whether to purchase comprehensive insurance for his vehicle for the year.

Without insurance, the possible outcomes over the year are:

Event	Probability	Cost to Operator (TZS '000)
No accident	0.75	0
Minor accident	0.20	-150
Major accident	0.05	-900

Insurance costs a flat **TZS 60,000** for the year and covers all accident costs completely (the operator pays nothing extra if an accident occurs).

- (a) Calculate the expected cost **without** insurance.
- (b) Calculate the expected cost **with** insurance. (*Hint: with insurance, the cost is always exactly TZS -60,000, regardless of what happens.*)
- (c) Based on expected value alone, which option has the lower expected cost?
- (d) Calculate $\text{SD}(X)$ for the uninsured case. What does this number tell the operator that the expected value in (a) does not?
- (e) Give one practical reason the operator might still choose to buy insurance, even if the expected cost is higher without it.